# Bibhabasu Bhattacharya

+91 6900671557 | bibhabasumalhar.jr@gmail.com | linkedin.com/in/BibhabasuBhattacharya | github.com/malhar02

## PROFESSIONAL SUMMARY

Electronics and Computer Engineering graduate (Thapar Institute of Engineering & Technology, 2026) with hands-on experience in Python, SQL, data analytics, and machine learning through applied projects and virtual industry programs. Skilled at translating data into actionable business insights, building end-to-end analytical pipelines, and communicating findings clearly to stakeholders. Strong problem-solving mindset, cross-functional collaboration experience through community leadership roles, and eagerness to contribute to digital business and technology-driven solutions at HCL.

## TECHNICAL SKILLS

- Programming & Query Languages: Python, SQL, C++
- Data Analytics & Machine Learning: Pandas, NumPy, Scikit-learn, PyTorch, Matplotlib, Seaborn, PCA, StandardScaler
- Databases & Tools: Oracle Live SQL, MySQL, Jupyter Notebook, Git/GitHub, VS Code, Arduino
- Embedded & IoT: Raspberry Pi, IoT-based Device Control, OCR Integration
- Cloud & DevOps**:** Microsoft Azure, Docker, Git/GitHub, Linux, CI/CD
- Other: Data Cleaning & Visualization, NLP & Regex-based Text Processing, OCR, Automation, Digital Literacy Training

## EXPERIENCE

**Business Analyst Intern — Xlearate Services, Noida** *January 2026 – June 2026*

- Worked in a hybrid business-and-technology role spanning client outreach, requirement gathering, marketing, and product development.
- Translated client business requirements into actionable inputs for the product/development team, gaining exposure to how requirements become digital solutions.
- Organized and maintained the company's service inventory and contributed to digital growth initiatives.

**BCG Data Science Job Simulation — Forage** *January 2026*

- Conducted customer churn analysis using Python (Pandas, NumPy) to clean, structure, and analyze business data.
- Built and optimized a Random Forest model achieving 85% accuracy, uncovering key churn drivers through data visualization.
- Presented actionable, data-driven insights and recommendations to senior leadership for strategic decision-making.

**Goldman Sachs Software Engineering Virtual Experience Program — Forage** *September 2025*

- Simulated an IT governance analyst role, evaluating security measures and identifying password vulnerabilities.
- Detected an outdated hashing algorithm by cracking passwords with Hashcat, demonstrating applied technical and security skills.
- Prepared a memo recommending stronger hashing algorithms and longer passwords to strengthen system security.

## PROJECTS

**Enterprise Intelligent Document Automation Platform** | *Python, FastAPI, OCR, LLM, PostgreSQL, Docker*

- Built an OCR + LLM pipeline to extract, classify, and validate logistics documents..
- Developed REST APIs and Dockerized services with logging, error handling, and health checks.
- Containerized services with Docker and designed a cloud-ready architecture for reliable deployment and troubleshooting

**AI Service Monitoring & Reliability Platform**| *Python, FastAPI, Docker, PostgreSQL, Azure*

- Developed a monitoring layer for AI/API services tracking availability, latency, error rates, request volume, and service health.
- Implemented health-check APIs, structured logging, exception handling, and automated alerts to identify service failures and SLA risks.
- Containerized services with Docker and designed a cloud-ready architecture for reliable deployment and troubleshooting.

**AI-based Address Detection & Validation (Capstone)** | *Python, NLP, Regex, NLTK, OCR, Raspberry Pi, Embedded Systems*

- Designed an NLP-driven pipeline for postal address extraction, parsing, and validation using Python and Regex.
- Integrated an OCR camera with Raspberry Pi (embedded hardware) for real-time text acquisition and preprocessing of physical documents.
- Deployed and tested the system on edge hardware, enabling lightweight, automated address verification.

## POSITIONS OF RESPONSIBILITY

**Co-Founder, Humans of Barak** *2022 – Present*

- Led social initiatives impacting 2,000+ beneficiaries and aided 800+ flood-affected families through municipal collaboration.
- Conducted digital literacy workshops for 150+ individuals, improving digital proficiency by 70%.

**Convenor, Mudra Society** *2026 – Present*

- Managed events for 800+ participants, coordinating cross-functional teams to ensure seamless operations.

## EDUCATION

**Thapar Institute of Engineering and Technology, Patiala** *Oct 2022 – May 2026*

- Bachelor of Engineering in Electronics and Computer Engineering